

Crop–Weed Segmentation Using RGB–NIR Images with U-Net

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Abstract—Accurate weed detection is essential for precision agriculture and automated weed control systems. In this study, a deep learning approach based on the U-Net architecture is implemented for pixel-level crop–weed segmentation using RGB and Near-Infrared (NIR) imagery from the Sugar Beets 2016 dataset. The model classifies each pixel into soil, crop, or weed classes. Experiments investigate the effects of data augmentation, hyperparameter tuning, and training stability across random seeds. The best configuration achieved approximately 0.60 Intersection over Union (IoU) and 0.97 pixel accuracy. Results demonstrate that incorporating NIR information improves vegetation detection and that proper hyperparameter tuning significantly improves segmentation performance.

I. INTRODUCTION

Weed detection is an important task in modern agriculture, enabling targeted herbicide application and reducing environmental impact. Traditional image processing methods struggle with varying illumination conditions and complex plant structures. Deep learning methods, particularly convolutional neural networks, have shown strong performance in semantic segmentation tasks.

In this work, we use a U-Net based segmentation model to distinguish crops, weeds, and soil from agricultural imagery. The model utilizes both RGB and Near-Infrared (NIR) channels, allowing it to exploit spectral information that helps differentiate vegetation from background.

The main goals of this study are:

- Train a semantic segmentation model using RGB+NIR imagery
- Evaluate the effect of different data augmentations
- Analyze the impact of hyperparameter tuning
- Examine training stability across random seeds

II. DATASET

The dataset used in this project is the **Sugar Beets 2016 dataset**, which contains agricultural field images captured with both RGB and Near-Infrared sensors.

Each sample consists of:

- RGB image (3 channels)
- NIR image (1 channel)
- Pixel-level annotation mask

The segmentation mask labels each pixel as:

- 0 – Soil / Background
- 1 – Crop

- 2 – Weed

To leverage all available spectral information, the RGB and NIR images are combined to form a 4-channel input image:

$$Input = [R, G, B, NIR] \quad (1)$$

This additional NIR channel helps distinguish vegetation from soil and improves segmentation quality.

III. MODEL ARCHITECTURE

The segmentation model used in this study is **U-Net**, a convolutional neural network architecture designed for semantic segmentation tasks.

U-Net consists of two main parts:

A. Encoder

The encoder extracts hierarchical features from the input image using repeated blocks of convolution, activation functions, and pooling operations. This stage reduces spatial resolution while increasing feature abstraction.

B. Decoder

The decoder reconstructs the segmentation mask through upsampling layers and convolution operations. Skip connections between encoder and decoder layers allow the model to retain spatial details that are important for precise segmentation.

The final layer produces a three-channel output corresponding to the three segmentation classes.

IV. TRAINING SETUP

The model was trained using the following configuration:

Parameter	Value
Input Channels	4 (RGB + NIR)
Output Classes	3
Loss Function	Cross Entropy Loss
Optimizer	Adam
Batch Size	8

A. Hardware Setup

Training was initially started on a local machine with an RTX 3060 GPU. However, training was relatively slow, with each epoch taking a considerable amount of time. To accelerate experimentation, the training process was later moved to Google Colab using a T4 GPU.

During the transition, the first prediction visualization appeared entirely purple. This occurred because the trained model weights were not correctly loaded after switching environments. Once the model was retrained properly in the Colab environment, the predictions matched the ground truth masks.

V. EVALUATION METRICS

Two evaluation metrics were used to assess model performance.

A. Pixel Accuracy

Pixel accuracy measures the percentage of correctly classified pixels:

$$\text{PixelAccuracy} = \frac{\text{CorrectPixels}}{\text{TotalPixels}} \quad (2)$$

B. Intersection over Union

Intersection over Union (IoU) measures the overlap between predicted and ground truth segmentation masks:

$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}} \quad (3)$$

IoU is commonly used in segmentation tasks as it penalizes both false positives and false negatives.

VI. DATA AUGMENTATION EXPERIMENTS

Several augmentation techniques were evaluated to determine their effect on model performance.

Augmentation	Pixel Accuracy	IoU
None	0.9709	0.5416
Horizontal Flip	0.9457	0.3373
Vertical Flip	0.9710	0.5413
Rotation	0.9685	0.5216
Brightness	0.9666	0.5103
Noise	0.9436	0.3145

Vertical flipping preserved performance, while horizontal flipping and noise significantly reduced IoU. This suggests that the spatial structure of vegetation plays an important role in segmentation.

VII. HYPERPARAMETER EXPERIMENTS

Different optimizers and learning rates were tested to improve model performance.

Increasing the learning rate to 5×10^{-4} improved the IoU significantly and produced the best overall performance.

Experiment	Pixel Accuracy	IoU
Baseline (Adam $1e^{-4}$)	0.9687	0.5246
Adam ($5e^{-4}$)	0.9757	0.5990
Adam ($5e^{-5}$)	0.9436	0.3145
SGD	0.9436	0.3145

Seed	IoU
0	0.595
1	0.586
2	0.575
3	0.585
4	0.579

VIII. RANDOM SEED STABILITY

To analyze training stability, the model was trained multiple times using different random seeds.

The mean IoU was approximately 0.584 with a standard deviation of 0.0067, indicating that the training process is stable and reproducible.

IX. QUALITATIVE RESULTS

Figure 1 presents an example segmentation output produced by the trained U-Net model. The first image shows the RGB input, the second image shows the ground truth annotation, and the third image shows the predicted segmentation mask.

The model successfully separates vegetation regions from the soil background. Most plant areas are correctly detected, demonstrating that the network has learned meaningful spatial and spectral features from the RGB–NIR input. However, the model does not clearly distinguish between crop and weed classes in all regions. In the prediction, vegetation areas are mostly assigned to a single class, while the crop class appears less frequently than in the ground truth.

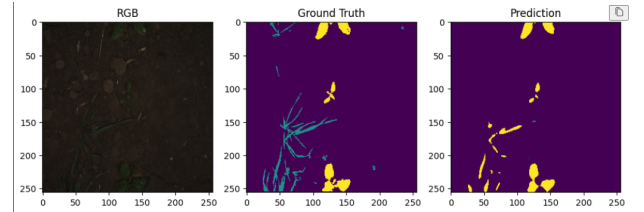


Fig. 1. Example segmentation result showing RGB input, ground truth mask, and predicted mask.

This behavior is likely caused by the visual similarity between crops and weeds as well as the class imbalance in the dataset. Since both classes represent vegetation and share similar spectral characteristics in RGB and NIR channels, the model tends to merge them into a single vegetation category. Additionally, resizing images to 256×256 may reduce fine structural details such as thin leaves, making class separation more challenging.

Despite these limitations, the model provides a reasonable segmentation of vegetation areas and demonstrates the effectiveness of U-Net for agricultural scene understanding.

X. CONCLUSION

This project presented a U-Net based semantic segmentation model for crop–weed classification using RGB and Near-Infrared (NIR) imagery from the Sugar Beets dataset. The input images were constructed by combining RGB and NIR channels to form a 4-channel input representation.

The final model configuration used in this study consists of a U-Net architecture trained with the Adam optimizer, a learning rate of 5×10^{-4} , a batch size of 8, and the Cross Entropy loss function. The input images were resized to 256×256 pixels to reduce computational cost and enable efficient training.

The experiments showed that:

- Incorporating NIR information improves vegetation detection.
- Hyperparameter tuning significantly affects segmentation performance.
- The training process is stable across multiple random seeds.

The final model achieved approximately 0.58–0.60 IoU and about 0.97 pixel accuracy on the test set. Although the model successfully separates vegetation from soil, distinguishing between crop and weed classes remains challenging due to their visual and spectral similarity.